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A truism in mammalian development is that fertilization is not difficult, but implantation is. About 70% of all human embryos die before birth, and the time of greatest embryonic loss is during implantation. (Cross et al., 1994). In this period, the embryonic, and extraembryonic yolk sac and placental cell lineages are determined and become essential for survival (Rappolee, 1999). In humans, failure to form a placenta properly leads to the clinically important conditions preeclampsia and intrauterine growth retardation (IUGR). Experiments have shown that perturbation of maternal or embryonic development only during the late preimplantation stages can lead to detrimental ramifications (Rutedge et al., 1994) during later development and at parturition (Kwong et al., 2000).
Hypothesis 1: Early cleavage division events are sensitive to gravity. Hypothesis 1a, the gravity-sensitive period is only during the 2-8-cell-stage when the block in embryonic development was observed (Kojima et al., 2000), and post-8-cell

Task Description:	development is normal (if embryos escape the block). Hypothesis 1b, the gravity-sensitive period also occurs during post-8-cell development after compaction. Specific aim 1. (1) Which stage of preimplantation embryonic development is gravity-sensitive? To test this, we will make daily measurements of several criteria of embryo development: (1) anabolic deviation (RNA, protein content and synthetic rates), (2) net growth rate (cell number), (3) components contributing to net growth rate (fraction in mitosis S phase, fraction in apoptosis), (4) morphogenetic events (compaction rate, cavitation rate), (5) and allocation to embryonic and placental lineages (fraction and total number inside and outside cells). Analysis of preimplantation development will be done to distinguish whether pre-compaction (2-8-cell-stage) development that is the primary period of perturbation (Kojima et al., 2000). What events are affected by microgravity? Zygotic genome activation (ZGA), compaction and cavitation will be tested for delay of initiation and quantitatively for relative strength. This will distinguish whether events during the 2-8-cell-stage block (Kojima et al., 2000) have an earlier onset. Post-compaction events such as cavitation and trophoblast lineage allocation will be tested for indication that development of the embryo is perturbed after the 2-8-cell-stage block-embryo. Hypothesis 2: Preimplantation embryos are sensitive to gravity and embryos cultured under conditions of microgravity during the (1) entire preimplantation development, (2) from 1-8-cell- stage, or (3) from post-compaction 8-cell-stage through blastocyst-stage show gravity-sensitive phenotypes. These phenotypes are observed in two post-implantation models; (1) when embryos from these three groups produce smaller litter sizes/weights after reimplantation and (2) when they produce deviant early post-implantation events measured in blastocyst or ICM outgrowth culture. Specific aim 2 (2) Are embryos that progress to the most advanced preimplantation stages (morula and blastocyst) in microgravity culture normal when reimplanted or cultured? (2a) Do embryos cultured in microgravity from (1) 1-cell-stage through blastocyst-stage, (2) from 8-cell-stage through blastocyst-stage, or (3) from 1-cell-stage through 8-cell-stage reimplant and produce normal quantity of newborns? Analyze number and weight of newborn mice produced from embryos culture in microgravity for parts or all of preimplantation development. If number or weight is decreased, do follow-up studies on size of embryos during mid- and late-uterine growth periods. (2b) Do blastocyst cultured in microgravity (1) from 1-cell to blastocyst-stage, or from 8-cell-stage through blastocyst-stage yield the normal placental/trophoblast and yolk sac/endoderm growth and differentiation events in blastocyst outgrowth culture at normal gravity? Analysis of blastocyst outgrowth culture using markers for trophoblast and endoderm proliferation, differentiation and apoptosis will be done to test for effects of microgravity on preimplantation embryos.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will inform us whether gravity has a role during mammalian embryogenesis as the embryo prepares to set up the head-to-tail, back-to-belly, and left-to-right axes. In addition, we will test whether embryonic movement set up during microgravity simulation is itself stressful and deleterious to embryogenesis before implantation into the uterus. In the first year of the grant, we have found that there is a profound and complete block of embryos during the step at which the first placental epithelium (and the first epithelium of any king)is made.
Task Progress:	Progress report, year three, NAG 2-1503: “Effects of microgravity on preimplantation mouse development” Abstract The most dangerous period of human embryonic development on earth is at implantation into the uterus. It is a general finding that fertilization occurs at a high rate, but that only about a third of fertilizations result in neonates. In humans, the greatest fraction of the 70% loss of post-fertilization embryos is around the time of implantation. Experiments in non-mammalian embryos show that gravity affects early development. Ground-based experiments based on centrifuge-generated forces show that gravity has a profound role in early reproduction on the formation of body axes in the chick and in amphibians. Recent spaceflight experiments suggest that gravity has effects during pregastrulation development in chicks, amphibians, and sea urchins. Recent data suggests that fertilization occurs normally under simulated microgravity, but that early embryonic cell divisions soon after are abnormal. It was recently shown that under microgravity conditions simulated by a clinostat, fertilization appeared normal when followed immediately by reimplantation into pseudopregnant females who were at normal gravity. Specifically, the number of newborns produced from in vitro fertilization (IVF) done under simulated microgravity was the same as at normal gravity. However, further experiments showed that if preimplantation culture continued in microgravity conditions, there were significantly fewer embryos progressing to the morula or blastocyst-stage. We wish to test whether microgravity disturbs embryonic events from fertilization through the 8-cell stage or from the 8-cell stage through formation of the blastocyst, prior to implantation. Embryonic development will be tested in vitro and then in vivo for effects of microgravity on development, cellular, molecular, and functional outcomes. We hypothesize that the microgravity-cultured embryos progressing to blastocyst may be normal, as the number of embryos stuck in 2-8-cell-stage block appears to account for the reported deficiency in numbers of embryos progressing to morula/blastocyst. At the 8-cell stage a crucial event called compaction occurs. Compaction requires the production of new mRNA transcripts from the zygotic genome (ZGA=zygotic genome activation). During this event the embryo makes the first epithelium. An alternate hypothesis is that post 8-cell-stage embryos are still gravity-sensitive after compaction. If this is the case embryonic events such as cavitation (making the cystic cavity in the blastocyst) and cell lineage allocation (producing cells that will make the embryo or the placenta), may be affected. Progress report Year two. We found that rotation of embryos in microdrop culture simulated a decrease in gravity to 0.1-3% normal gravity (MGS) when rotated at 7.5 rotations/minute. Next, we tested whether the conditions of simulation of gravity were stressful to the embryos and might add an additional variable besides that of simulated microgravity. We did this because Dr Hammond had found that microgravity simulation in the RCCS-1 causes turbulent stresses that may lead to stress on cell [or embryos]. Since no stress measurements have been made on embryos we used stress activated protein kinase (SAPK) and p38MAPK as markers of stress induction in embryos. In each case, high salt (hyperosmolarity) caused the phosphorylation/activation of SAPK and p38MAPK. In agreement with the literature, this phosphorylation causes a distribution of the SAPK to the nucleus of the embryo. In contrast, significantly lower amounts of SAPK were detected in the nucleus of cells in preimplantation mouse embryos after 24hr culture in microdrops in the RCCS-1 bioreactor (at simulated microgravity of about 0.1% normal gravity; p=0.03 t test). Culture of embryos in RCCS-1 or in static culture produced equivalent amounts of nuclear localized SAPK phospho after 24hour culture. The embryonic

	stage tested was E3.5 to E4.5. This would approximate 30-60 cell-stage embryo. During this time period of embryonic development, no adverse effects on embryonic morphogenesis or cell number were noted, but there was a delay in development in MGS and to a lesser extent in rotational control (RC) compared with static control (SC). During the recovery period, there was a further deterioration in MGS>>RC>>>>SC Year three. Dr Wang concluded her study by finding that E1.5 (2-cell stage) was as sensitive to MGS caused lethality as E3.5, but that E2.5 (8-cell stage) was the most sensitive to MGS-induced lethality in MGS>RC>>>>SC. The 8-cell stage was the most sensitive time period of development, but RC was also very lethal at this stage. This is the period when the first placental epithelium forms and is known to be an important developmental stage. Dr Wang left the lab and Dr Xie took over the project. Dr Xie established an RCCS1 rotation similar to Dr Wang's, but used larger aqueous media droplets that did not orbit in the oil as the vessel rotated at 7.5 RPM. This created more shear stress and minimal MGS. E3.5 embryos became apoptotic and died within 12hr or MGS and within 24hr of RC, suggesting a more rapid lethality than orbital MGS that approximated 0.1% gravity. SAPK/JNK phosphorylation preceded the TUNEL and rapid lethality of MGS and slower lethality of the RC group. A SAPK/JNK inhibitor significantly reduced the fraction of TUNEL cells, suggesting that SAPK/JNK was causal in. Fos was quickly induced (0.5hr MGS), suggesting shear stress developed rapidly and was sensed by the outer placental epithelial cells. In 1970, Bronson and McLaren reported that removing the Zona pelucida (an acellular 4% protein coat around the oocyte preimplantation embryo) at the 8-cell stage was lethal to embryos that were reimplanted into the oviduct where they would move, but not in static embryos in culture. We tested whether the Zona pellucida might protect the embryo from shear stress. Embryos with the Zona removed were significantly more susceptible to lethality induced by shear stress, but embryos without Zona were not significantly more TUNEL in static culture than embryos with Zona. Summary; MGS or shear stress leads to SAPK/JNK phosphorylation/activation that precedes lethality. SAPK/JNK inhibitor (SP600125) can largely block shear stress-induced TUNEL/apoptosis, suggesting that SAPK activity is causal for apoptosis. We are currently quantitating the amount of shear stress in the non-orbital mode of MGS, and testing for the interaction of shear stress and SAPK/JNK activation of JunC phospho at 0.5hr to find if jun:fos heterodimers may mediate early preparation for TUNEL/apoptosis. This work was supported by the following grants NASA (NAG2-1503), NIH (R01 HD40972A). Additional help was kindly given by Dr Josh Zimmerberg of NICHD, who provided 2 additional RCCS1 and 10mL HARVS.

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